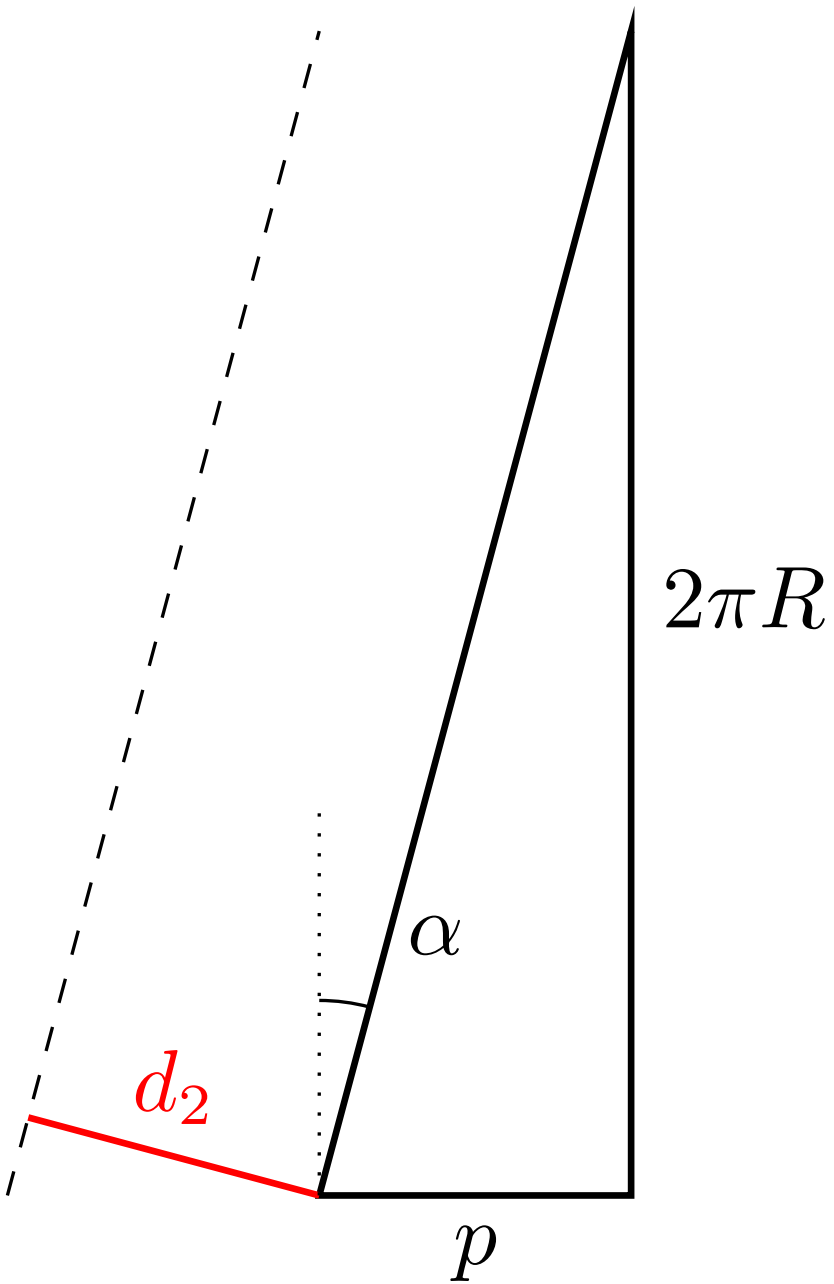

$$\alpha = \phi/2$$

$$d_2 = p\cos\alpha$$

$$2\pi R = p\cot\alpha$$